Grade 7 Computer Science Solved Exercises - Balochistan Textbook Board

Balochistan Textbook Board (BTBB)

Unit 1: ICT Fundamentals

Exercise (pages 20-22): Tick the correct choice for the following statements/questions:

i. There are _____ types of communication ports in a computer.

- ✓ 2
- 8
- 10
- 16

ii. Data is transferred in or out using more than one wires is called _____.

- serial
- ✓ parallel
- band width
- networking

iii. USB stands for _____.

- Unique Serial Bus
- Universal Solo Bus
- Universal Sort Bus
- ✓ Universal Serial Bus

iv. The Latest USB is _____.

- Type A
- Type B

- ✓ Type C
- Type D

v. HDMI stands for _____.

- ✓ High-Definition Media interface
- High Data Media Interface
- High-Definition Mode Interface
- High Data Mode Interface

vi. _____ is the point that connects the peripheral to the mother board.

- ✓ PCI
- USB
- Modem
- NIC

vii. _____ memory loses its contents when power off.

- ✓ RAM
- ROM
- USB
- Buffer

viii. RAM is a type of _____.

- ✓ Volatile memory
- Peripheral
- ROM
- PROM

ix. _____ Port is used to install Network card.

- AGP
- ISA
- AMR
- ✓ PCI

x. Two or more connected computers are called _____.

- ✓ Network
- Automation
- Multimedia
- System

Note on MCQ i: the unit names two common types of computer ports, Serial Ports and Parallel Ports (section 1.4.3), so 2 is the intended option. Note on MCQ vi: the unit calls the connection point an expansion slot or bus slot; PCI is the slot type among the options. Note on MCQ ix: the unit lists PCI as used for the Network card (section 1.5.2).

Exercise: Fill in the blanks.

i. Robots are machines that can be _____ to perform tasks autonomously or semi-autonomously.

programmed

ii. A computer port is also called _____.

a jack or socket

iii. A _____ is an interface through which transmission of data is in one bit at a time over a single communication line.

Serial port

iv. _____ is the common standard and supports a data rate of 400 Mbps.

USB 3.0

v. USB port can be used to transfer data, connect peripherals and even act as power supply for _____ connected to it.

devices

vi. HDMI is an abbreviation of _____.

High-Definition Media Interface

vii. PCI Express (PCIe) is short for _____.

peripheral component interconnect express

viii. A _____ is any device that connects to the computing unit but is not part of the computing unit.

peripheral

ix. _____ refers to the physical components within a computer that store data and instructions, either temporarily or permanently.

Computer memory

x. _____ is a type of non-volatile memory, meaning it retains its data even when the computer is powered off.

ROM

Note on fill ii: the unit defines a computer port as a jack or socket used for communication between the computer and its peripheral devices (section 1.4.3). Note on fill iv: the unit writes the rate as 400 MBps for USB 3.0; the exercise prints 400 Mbps.

Exercise: Answer the following questions.

i. Define 3D Printing.

3D printing, also known as additive manufacturing, is a method of creating a three-dimensional object layer-by-layer using a computer-created design. Instead of ink, 3D printers use materials like plastic, metal or resin to create solid objects.

ii. Define AI.

Artificial Intelligence (AI) is a branch of computer science that uses algorithms, computational power and data to generate machines that can achieve tasks like human intelligence, such as visual sensitivity, speech recognition, translating written and spoken language, decision making, learning, problem solving, planning, reasoning and classifying patterns.

iii. What is Machine Learning?

Machine Learning (ML) is a branch of artificial intelligence that uses algorithms to learn from data and improve the performance of artificial intelligence, such as predicting outcomes and the future. For example, a model that learns from the inputs 2, 3 and 4 giving outputs 4, 9 and 16 can predict that the output of 5 will be 25, because the outputs are the squares of the inputs.

iv. Define Virtual reality.

Virtual Reality (VR) is a technology that creates a simulated environment, allowing users to experience and interact with a digital world as if they were physically present in it. This immersive experience is achieved by using a VR headset, which covers the user's eyes and often includes sensors to track head movements and hand controllers.

v. What is the difference between AI (Artificial Intelligence) and ML (Machine Learning)?

AI is the wider branch of computer science that makes machines able to perform tasks like human intelligence, such as decision making, reasoning, speech recognition and problem solving. ML is a branch of AI: it is the part that uses algorithms to learn from data and improve performance, for example by predicting outcomes. So every machine learning system is an AI system, but AI also includes systems that do not learn from data.

vi. Define Networking.

A network is a system that contains two or more computers connected together. They are connected through cables, telephone lines, radio waves, satellites or infrared light beams, and share their resources like hard disks, printers and CDs, exchange files, or allow electronic communication. The devices or computers connected to a network are called nodes.

vii. Write any three advantages of Networks.

Three advantages of a computer network from the unit are: (1) File Sharing: with the help of a computer network, we can easily share

data between different users. (2) Hardware Sharing: all computers in the network can share resources such as printers, scanners, CD-ROMs, modems and hard disks. (3) Application Sharing: networks allow their users to access the same application at the same time, so many users can work on the same application from anywhere in the world.

viii. What is the difference between RAM and ROM?

RAM (Random Access Memory) is a type of volatile memory: it only stores data while the computer is powered on, temporarily holds the data and instructions the CPU needs to access quickly, and once the computer is turned off, all data in RAM is lost. ROM (Read-Only Memory) is a type of non-volatile memory: it retains its data even when the computer is powered off, stores crucial data and instructions required for the computer to boot up, like the BIOS (Basic Input/Output System), and unlike RAM, data in ROM cannot be easily modified or erased, which is why it is called read-only.

ix. What is the difference between Volatile memory and non-volatile memory?

Volatile memory only stores data while the computer is powered on and loses all its contents when the power goes off; RAM is the unit's example. Non-volatile memory retains its data even when the computer is powered off; ROM is the unit's example, and it holds the instructions needed to boot the computer.

x. Describe computer peripherals.

A computer peripheral is any device that connects to the

computing unit but is not part of the computing unit. The computing unit consists of the central processing unit (CPU), the motherboard and the power supply. Peripherals are detachable and can be replaced; for example, the hard drive is a peripheral. There are three types of peripherals: input devices, output devices, and input/output and storage devices. Section 1.6 adds that peripheral devices are external devices connected to a computer that add functionality or enable users to interact with the system, and classifies them into input devices, output devices and storage devices.

Exercise: Label the diagram.

a. The first port (rectangular, 4-pin) is:

USB Type-A.

b. The second port (square) is:

USB Type-B.

c. The third port (small oval) is:

USB Type-C.

d. The fourth port (wide trapezoid) is:

USB Mini.

e. The fifth port (small trapezoid) is:

USB Micro.

The five labelled ports are the unit's Figure 1.4.4a set: USB-A,

USB-B, USB-C, USB Mini and USB Micro. The unit states there are four kinds of USB ports, Type A, Type B or mini-USB, Type C and Micro USB, and lists USB Type-C as the latest USB, supposed to replace Types A and B.

Lab Activities

Guidance (Lab Activity 1): list the peripherals attached to your own computer, for example keyboard, mouse, monitor, printer and speakers; the book fixes no single list, since it depends on the student's own computer. Guidance (Lab Activity 2): visit your school computer lab and identify the network devices found there, for example the router, switch and network cables; the book fixes no single answer. Guidance (Lab Activity 3): with your teacher's help, remove all the cables attached to a computer system and attach them again, then switch the computer ON; this is a hands-on task with no written answer in the book.

Unit 2: Word Processing

Exercise (pages 43-45): Tick the correct choice for the following statements/questions:

i. Microsoft Word is a _____.

- language
- ✓ word processor
- utility program
- spread sheet

ii. To open document, press _____.

- F1
- Alt+F4
- Ctrl + P
- ✓ Ctrl + O

iii. _____ is the short cut key to print a file.

- F1
- Alt+F4
- ✓ Ctrl + P
- Ctrl + S

iv. A _____ is graphical representation of text.

- ✓ font
- bullets
- indentation

design

v. _____ contain information such as page numbers, title and date.

✓ Header/Footer

Title

Quick tool bar

Ribbon

vi. _____ is the shortcut key to check spellings.

✓ F7

Alt+F4

Ctrl + P

Ctrl + S

vii. _____ is used to cancel our last action in MS Word.

F1

Alt+F4

Ctrl + P

✓ Ctrl + Z

viii. Short cut key to Redo our last action is _____.

✓ Ctrl+Y

Alt+F4

Ctrl + P

Ctrl + S

ix. _____ is a number that indicates how many points are in

height of the text.

font

✓ font size

indentation

face type

x. _____ is used to save a document first time.

Import

✓ save

save as

export

Note on MCQ x: the unit states the Save command is used to save a file by giving a file name and location for the first time, while Save As creates a copy of a document while keeping the original.

Exercise: Fill in the blanks.

i. Microsoft Office is a family of software developed by _____.

Microsoft

ii. Microsoft Word, Excel, PowerPoint are called _____.

Applications

iii. The _____ is a horizontal bar located at the top of a window.

Title bar

iv. Press _____ on your keyboard to paste text, shape or image.

Ctrl + V

v. To open a file, press _____.

Ctrl + O

Exercise: Answer the following questions.

i. Define Word processor.

A word processor is a computer program used to create, modify text, format text, apply different operations on text and print text documents. Its key advantage is its ability to make changes easily, such as correcting spelling, adding, deleting and formatting text.

ii. What is the difference between Cut, Copy and paste?

Copying: when you copy text or an object in MS Word, you make a duplicate of the selected content without removing it from its original place; the copied content is stored temporarily in the clipboard, allowing you to paste it elsewhere in the document or in another document. Cutting: when you cut text or an object, it is removed from its original location and stored in the clipboard, allowing you to move it to a new location; cutting is useful when you want to relocate content rather than duplicate it. Pasting: pasting places the clipboard content at the insertion point, using the Paste command on the Home tab or Ctrl+V on the keyboard. Shortcuts: Cut Ctrl+X, Copy Ctrl+C, Paste Ctrl+V.

iii. What is the difference between Undo and Redo?

Undo cancels your most recent action when you make a mistake, for example accidentally deleted text: select the Undo command on the Quick Access Toolbar or press Ctrl+Z, and you can continue using the command to undo multiple changes. Redo reverses the

last undo, restoring the action you undid: use the Redo command or press Ctrl+Y on your keyboard.

iv. What is the difference between Header and Footer?

Header and Footer both contain information such as page numbers, title and date. The Header is the area at the top of the page and the Footer is the area at the bottom of the page; both repeat on the pages of the document. To add either: click the Insert tab on the Ribbon, click Header or Footer, and choose what you want to place there (date, time, page number).

v. What is the difference between Save and Save As?

Save: the Save command is used to save a file by giving a file name and location for the first time. Save As: the Save As command is used to create a copy of a document while keeping the original; when you use Save As, you need to choose a different name and/or location for the copied version. To save a document: select the Quick Access Toolbar Save command (or press Ctrl+S), choose where to save the file, enter a file name, then click Save.

vi. How to print a document?

To print a document: (1) Open the document you want to print. (2) Click the File tab followed by the Print option, which displays a preview of the document in the right column, or press Ctrl+P. (3) Select from the print options according to your preferences: number of copies, print custom range (a particular page), print one sided or on both sides, collated or uncollated, orientation (Portrait by default, or Landscape), page size (A4 by default), custom

margins, and pages per sheet (1 by default). (4) Once you are done with your settings, click on the Print button, which sends your document to the printer for printing.

vii. How to write anything in a shape.

To write in a shape: (1) Click the Insert tab on the Ribbon. (2) Click the Shapes icon on the Illustrations group. (3) Click the shape you want, for example Rectangle, and drag the mouse to draw the shape on the page. (4) Right-click inside the shape and select the Add text option, then type your text inside the shape.

viii. What is the difference between Bold and Italic?

Bold makes the selected text thick and dark so it stands out; its shortcut is Ctrl+B. Italic slants the selected text to the side; its shortcut is Ctrl+I. To apply either: select the text you want to change, then select the Bold, Italics or Underline button in the Home tab, or use the shortcut keys. Underline (Ctrl+U) draws a line under the selected text.

ix. Write the procedure to add header and footer in your document.

To add a header and footer in your document: (1) Click the Insert tab on the Ribbon. (2) Click Header or Footer. (3) Choose what you want to place on the Header or Footer, such as date, time or page number. Headers and Footers contain information such as page numbers, title and date.

x. Describe indentation in a paragraph.

Indenting paragraphs allows you to set text within a paragraph at

different margins. The unit's options are: First Line, which controls the left boundary for the first line of a paragraph; Hanging, which controls the left boundary of every line in a paragraph except the first; One Left (left indent), which controls the left boundary for every line in a paragraph; and Right, which controls the right boundary for every line in a paragraph. To indent paragraphs: click the Indent buttons repeatedly to increase the size of the indent, or click the dialog box launcher of the Paragraph group, click the Indents and Spacing tab, and select your indents.

Exercise: Label the diagram.

1. The band of tabs and command groups is the:

Ribbon: the toolbar at the top of the window that contains various commands and tools, organized into tabs, each with its own set of tools.

2. The small toolbar at the top left (AutoSave, Save, Undo, Redo) is the:

Quick Access Toolbar: located at the top left of the MS Word window; it gives quick access to frequently used features, such as Save and Undo.

3. The bar showing Document2 - Word is the:

Title bar: a horizontal bar located at the top of a window; it displays the title of the software and the name of the current document, here Document2.

4. The box labelled Search at the top is the:

Search box: type here to search for commands, help and templates in Word.

5. The purple circle icon at the top right is the:

Account (user profile) icon: the signed-in user's account picture at the top right of the window.

6. The X at the top right corner is the:

Close button: the X at the top right corner closes the Word window; beside it are the Minimize and Maximize buttons.

7. The marked strip at the left edge of the page is the:

Vertical ruler: the ruler displayed at the left-hand side of the window; rulers are used to measure the dimensions of the document.

8. The mark at the top left of the text area is the:

Insertion point (cursor): the flashing cursor at the top left of the text boundary area (page) where typed text appears.

9. The large white area is the:

Blank page (document area): the blank page to write on.

10. The numbered strip above the page is the:

Ruler (horizontal ruler): the ruler displayed at the top of the page; rulers are used to measure the dimensions of the document.

11. The bar at the right edge of the window is the:

Vertical scroll bar: the bar at the right edge of the window used to scroll the document up and down.

12. The slider at the bottom right (150 percent) is the:
Zoom slider: at the bottom right of the status bar; drag it to change the zoom level, shown here as 150 percent.

13. The small view icons at the bottom right of the status bar are the:
View buttons: the view controls at the bottom right of the status bar, for switching between Read Mode, Print Layout and Web Layout views.

14. The bar at the bottom showing Page 1 of 1 and 0 words is the:
Status bar: the bar at the bottom of the window showing document information such as Page 1 of 1, word count, language and accessibility status.

Note on labels 13 and 14: both are parts of the status bar. Label 13's arrow ends at the view-layout icons at the bottom right, and label 14's arrow ends at the document-information area (page, words, language, accessibility) of the same bar; they are labelled as view buttons and status bar (document information) respectively, following the unit's Figure 2.2.3b labels (Status bar, Document info, Print Layout, Zoom).
Exercise: Complete the shortcut-key table.
Print a document:
Ctrl + P.

Italic highlighted selection:

Ctrl + I.

Underline the selected text:

Ctrl + U.

Close:

Alt + F4.

Copy:

Ctrl + C.

Paste:

Ctrl + V.

Bold:

Ctrl + B.

Note on Close: the unit's shortcut tables give Ctrl+S for Save, Ctrl+O for Open, Ctrl+X for Cut, Ctrl+C for Copy, Ctrl+V for Paste, Ctrl+B for Bold, Ctrl+I for Italic and Ctrl+U for Underline; Alt+F4 is the unit's MCQ option for closing a window (Unit 2 MCQ ii/vi/vii options) and the standard Windows close shortcut, used here for the table's Close row.

Lab Activities

Guidance (Practical 1): type the given paragraph in Calibri font, size 12; change the font type to Lucida Calligraphy and size 14; set

alignment to the right margin; save the file as ms-word.doc in the My Documents folder; open the file again; save the file with the new name my_file in the D: drive; take a printout of the document. This is a hands-on task; the book fixes no single written answer. Guidance (Practical 2): write an application for one day's leave in MS Word, add proper indentation and font, and print the application; the wording of the application is the student's own. Guidance (Practical 3): create the given table with Insert Table: columns Student Name, Marks Obtained, Total Marks, Remarks, with rows Student 1 (430/500), Student 2 (470/500), Student 3 (485/500), each remarked Promoted to next class. Guidance (Practical 4): create the given While loop flowchart: While loop Start, then Test Condition? (True leads to Execute loop body, which loops back; False leads to While loop ends). These are hands-on tasks with no single written answer in the book.

Unit 3: Multimedia Presentation

Exercise (pages 58-59): Tick the correct choice for the following statements/questions:

i. How to Run presentation in PowerPoint _____?

✓ Using slide show

Using animation

Using transition

using format Tab

ii. Which feature is not in MS PowerPoint _____?

Save

✓ scan a virus

Shape

Print

iii. What is shortcut key to run Presentation in Power Point?

Ctrl + V

Ctrl+ C

✓ F5 key

Ctrl + A

iv. Adding Image in slide we must use?

✓ Insert Picture

Hyperlink

Layout Tab

F5 Key

v. Which of the following file formats can be added to a PowerPoint show?

jpg

Bmp

wav

✓ All of these

vi. Which Menu Tab we should use to add animation

Insert Tab

Slide Show Tab

✓ Animation Tab

design Tab

vii. Effect that occurs when you move from one slide to the next slide is called

Animation

Slide show

✓ transition

Design

viii. In which bar we can see the current position of the slide?

ribbon

title bar

✓ status bar

view option bar

ix. Press _____ to end presentation.

- Tab
- ✓ Esc
- E
- P

x. An Animation which controls how an object appears on the slide. For example, text might fade in is called

- Vertical
- Portrait
- Landscape
- ✓ entrance

Exercise: Fill in the blanks.

i. A _____ animation controls how an object appears on the slide. For example, text might fade in, or an image might slide in from the side.

Entrance

ii. An _____ animation controls how an object leaves the slide. For example, text might fade out, or an image might fly off the slide.

Exit

iii. _____ make a presentation non-linear by connecting it with other slides, documents and web pages.

Hyperlinks

iv. Anything on the slide is called _____.

object

v. _____ are visual effects which make your texts, images, shapes, or charts come 'alive.'

Animations

Exercise: Answer the following questions.

i. Define PowerPoint application.

PowerPoint is a complete presentation program/application developed by Microsoft. PowerPoint offers word processing, drawing, graphing and presentation management tools.

ii. Define Multimedia.

A multimedia presentation is a type of presentation that uses different types of media like text, images, videos, sounds and animations to share information in a more engaging way. Instead of just using words, it combines these elements to help explain ideas more clearly and make the presentation more interesting. Its main components are: Text (words, sentences, paragraphs); Images (pictures, photos, graphics); Audio (sound effects, music, spoken words); Video (moving images and sound); Animations (moving graphics or illustrations); and Interactivity (clicking on links, answering questions, navigating through sections).

iii. Define Animation.

Animations are visual effects which make your texts, images, shapes, or charts come 'alive.' Animation is the visual effect that

adds movement and sound effects on a PowerPoint slide, such as on the text, image, shape or tables. The four types of animations are Entrance (how the object enters the slide), Emphasis (effects that occur by a mouse click, such as spinning an object), Exit (how the object exits the slide) and Motion Paths (the object moves within the slide along a predetermined path, like a circle).

iv. Define Transition.

Transitions are the animation-like effects used in PowerPoint presentations to remove one slide from the screen and display the next slide. A transition is therefore the effect that occurs when you move from one slide to the next slide, while an animation acts on objects within a single slide.

v. Define slide and slide show.

A slide is a single page of a presentation; a presentation is made up of multiple slides, and by default a new presentation begins with one slide. A slide show is the running of the presentation, showing the slides full-screen in sequence.

vi. How can we Run our presentation?

Run your slide by clicking on the slide show button on the right-most bottom of the window, or press F5. Press Esc to end the presentation.

vii. How to add an audio on the slide?

To add audio on the slide: (1) Click on the Insert Menu. (2) Select Audio in the Media tab. (3) Click on Audio on My PC. (4) Select the

Audio/MP3 from your computer.

viii. How to add a picture on the slide?

To add a picture on the slide: (1) Click on the Insert Menu. (2) Click on the Picture icon. (3) Click on This Device. (4) Select the image from your computer and adjust it on the slide.

ix. How to add a video clip on the slide?

To add a video clip on the slide: (1) Add a new slide. (2) Click on the Insert Menu. (3) Select Video in the Media tab. (4) Click on This Device. (5) Select the video from your computer.

x. How to use Hyperlink on the slide?

Hyperlinks make a presentation non-linear by connecting it with other slides, documents and web pages. To add a hyperlink on text: (1) Create the presentation. (2) Select the text, right-click the selected text, then click the link, or click on the Insert tab and click the Link command; the Insert Hyperlink dialog box will open. (3) Select 'Place in This Document' in the link-to tab and click the target slide from the right option window. The same method applies to shapes, video, web pages and MS Word document files.

Exercise: Label the following diagram

Arrow at the top left corner points to the:

Quick Access Toolbar: the small toolbar at the top left of the PowerPoint window.

Arrow at the left edge, at the tab row, points to the:

File tab (Backstage view): the first tab at the left of the Ribbon.

Arrow at the lower left, pointing at the slide thumbnail, points to the:
Slide: the slide thumbnail in the Slides Pane at the left.

Arrow at the bottom left, pointing at the notes area, points to the:
Notes pane: the Click to add notes area below the slide, where speaker notes are written.

Arrow at the top right corner points to the:
Window controls: the Minimize, Maximize and Close buttons at the top right corner of the window.

Long arrow from the right, pointing at the top bar, points to the:
Title bar: the bar showing the presentation name, Presentation1 - PowerPoint.

Arrow at the bottom right, pointing at the large white slide, points to the:
Slide: the main slide editing area with the Click to add title and Click to add subtitle placeholders.

Note on the diagram: the book's arrows are unnumbered, so the labels follow the arrow endpoints in order around the window and the unit's own Figure 3.2b vocabulary (File tab, Quick Access Toolbar, Title Bar, The Ribbon, Slide, Slides Pane, Status Bar, Placeholders). The top-right arrow is read as the window controls

and the bottom-left arrow as the notes pane; both are best-readings from the arrow endpoints.

Lab Activities

Guidance (Practical 1, Part One: Creating a Title Slide): double-click the PowerPoint icon on the desktop; PowerPoint opens with a single blank slide with the Title Slide selected; click once in the Click to add title text box and type World of Travel; click once in the Click to add subtitle text box and type Take a trip; click in a clear place on the slide to deselect the two text boxes. Guidance (Practical 2, Part Two: Creating Slides Using Layouts): on the Home tab, in the Slides group, click the top half of the New Slide button to insert a new slide with the Title and Content layout; title Finding Travel Online; content Try a variety of websites, then Enter, Tab one level in and type Known travel sites, Enter, Search engines, Enter, Newsgroups. Guidance (Practical 3, Part Three: Creating a Slide with a different layout): click the bottom half of the New Slide button, choose Two Content; title Some Popular Travel Sites; left content Kayak, Expedia, Priceline; right content Travelocity, Orbitz, Hotwire. Guidance (Practical 4, Part Four: Creating another slide with the same layout): click the top half of New Slide to insert a slide with the same layout; title Have A Safe Trip!; left content Don't Forget..., then Enter and Tab one level in and type Money!, Enter, Passport!, Enter, Emergency Numbers!, Enter, Medication!. These are hands-on tasks; the book fixes no single written answer beyond the steps and texts given.

Unit 4: Electronic Mail

Exercise (pages 73-74): Tick the correct choice for the following statements/questions.

i. Email stands for _____.

- ✓ Electronic mail
- Emergency mail
- Embedded mail
- Electrical mail

ii. _____ is a set of secret characters or words used for authentication.

- Inbox
- deleted
- outbox
- ✓ password

iii. The _____ is the default folder for all incoming mails.

- RAM
- SSD
- ✓ inbox
- outbox

iv. All unwanted, adds etc., mails are copied in _____.

- inbox
- sent

- ✓ junk
- delete

v. Emails having any issue goes in _____.

- ✓ outbox
- RAM
- ROM
- Junk

vi. The process of sending messages directly form one computer to another is called

- ✓ Email
- information
- program
- hard copy

vii. The fastest communication source is

- portal
- FedEx
- ✓ Email
- Internet

viii. E-mail is received in _____ even your computer is off.

- portal
- ✓ server
- memory
- hard disk

ix. After checking email, you should _____.

- login
- ✓ logout
- shutdown
- restart

x. To send a picture with email, we have to _____ the file.

- scanner
- digital camera
- biometric sensor
- ✓ attach

Note on MCQ vii: the unit lists Time Saving among the advantages of email, email usually reaches its destination within seconds, and email can be received even if the recipient's computer is off because it is stored on the servers; Internet is the network over which email travels, while Email is the communication service named here.

Exercise: Fill in the blanks.

i. _____ is the short of electronic mail.

E-mail

ii. Each user of email is assigned a unique name, called _____.

E-mail address

iii. In the email address @ is called _____.

@ (at) symbol

iv. The first part of email address is known as _____.

username

v. The next part of @ in the email address is known as _____.

domain name

Exercise: Define the following terms.

i. Inbox

Inbox is a special folder in your email account. The Inbox is the default location for all incoming mail. Emails are arranged with respect to Date and Time. You can read a message by clicking on Inbox and then clicking on a particular message.

ii. Email

Email (electronic mail) is a way to send and receive messages and files (text, pictures and sounds etc.) across the Internet. To send and receive email, you must have an e-mail account, be connected to the internet, and have the e-mail address of those to whom you want to send email.

iii. Junk mail

Junk is a separate folder for messages which are not meant for us. Junk emails include advertisements, jokes or any harmful messages. The unit also notes it is important to delete junk mails.

iv. Sent

The Sent folder stores all the emails that we have sent to other people.

v. Outbox

The Outbox is a temporary folder that holds your outgoing emails if there are any issues sending the message until they can be sent.

vi. CC

CC stands for Carbon Copy. It includes those recipient addresses whom we want to keep informed but who are not exactly the intended recipient. BCC stands for Blind Carbon Copy: when you send an email as a BCC, the recipients do not know that a copy has been sent to other people.

Exercise: Differentiate the followings.

i. Inbox and Outbox

Inbox is the default folder for all incoming mail, arranged by Date and Time; it holds the messages you receive. Outbox is a temporary folder that holds your outgoing emails if there are any issues sending the message until they can be sent; it holds messages on their way out, not messages you received.

ii. Login and Logout

Login (sign in) is entering your email account with your email address and password, so the email program can be configured with your account information and you can access your mailbox. Logout (sign out) means to close your email account; after checking, composing and replying to emails we sign out of our email account, using the Sign out button which appears on the top right side of the web page.

iii. Deleted and Junk

Deleted is the folder where all deleted emails are sent; they remain there for 30 days, and after 30 days all deleted files are removed permanently. Junk is a separate folder for messages which are not meant for us, including advertisements, jokes or harmful messages; junk mails arrive on their own and are not mails you deleted yourself.

iv. Username and domain name

Username is the unique (user) name within the domain, the first part of an email address before the @ symbol, for example uniquename in uniquename@domain.com. Domain name is the email provider's domain, the part after the @ symbol, for example domain.com, where the dot separates parts of the address and com is the top-level domain name.

Exercise: Write short answers to the following questions.

i. What is meant by email?

Email (electronic mail) is a way to send and receive messages and files (text, pictures and sounds etc.) across the Internet. To send and receive email you must have an e-mail account, be connected to the internet, and have the e-mail address of those to whom you want to send email.

ii. What are the advantages of email?

The unit's advantages of email are: (1) Time Saving: email usually reaches its destination within seconds. (2) Cost Effective: email is

the cheapest way of transferring messages from one place to another; it is as low as free. (3) Attachments: emails allow you to send various types of files as attachments, such as documents, photos and videos, making it easy to share information. (4) No Disturbance During the Work: the receiver is not interrupted by the arrival of the email; it is stored in the mailbox and can be opened later. (5) Convenience: email can be received even if the computer of the recipient is off; email is stored in the servers and can be opened later. (6) Insured Delivery: since every email has the address of the destination, it is certain that it will reach the destination. (7) One Email to Many: one email can be sent to many persons just by adding all the email addresses and clicking Send. (8) Auto Response: it is possible to send automated emails with a certain text, for example to tell the sender that you are busy or on vacation. (9) Email Never be Anonymous: the address of the sender is automatically attached to the sent email.

iii. What is the function of email address?

An email address is the unique name assigned to each user for his email account. It is the address to which messages are sent and by which the email service collects a message into the recipient's mailbox. Email addresses are always written in a standard format that includes a username, the @ (at) symbol, and the email provider's domain, for example uniquename@domain.com.

iv. Describe the procedure for attaching a file to an email message.

To attach a file to an email message: (1) Open your email account. (2) Click on the Compose button. (3) Write the recipient email

address and write your message. (4) Click on the attachment (paperclip) icon. (5) Select the file from your computer which you want to send. (6) Click on the Send button. An attachment provides more information with the email; it could be a document for review, a picture to share or any other file type.

v. Why we mention the subject of an email message?

The Subject field indicates the purpose of the e-mail. It should be precise and to the point, so the recipient knows what the message is about before opening it. The Subject is one of the header fields of an email message, along with From, Date, To, CC and BCC.

vi. Where the deleted mails can be found?

All deleted emails are sent to the delete folder. They remain there for 30 days, and after 30 days all deleted files are removed permanently.

vii. What is the use of outbox?

The Outbox is a temporary folder that holds your outgoing emails if there are any issues sending the message until they can be sent. It is the use of the Outbox to hold outgoing mail that could not be sent yet, rather than losing it.

viii. What is contact list?

A contact list is a list of people or organizations for instant messaging and email. A contact list shows screen names that represent actual people for communicating or sharing information. To find your Gmail contacts: open Gmail on your PC in a web

browser, click the Google apps button at the top-right next to your account icon, and click Contacts; Contacts will then launch, and you can use it to add, find, edit and manage your contacts.

ix. What is the use of CC?

CC stands for Carbon Copy. Its use is to include those recipient addresses whom we want to keep informed but who are not exactly the intended recipient; all recipients can see the CC addresses.

x. What is the use of BCC?

BCC stands for Blind Carbon Copy. Its use is to send a copy of an email to recipients without the other recipients knowing that a copy has been sent to them; when you send an email as a BCC, the recipients do not know that a copy has been sent to other people.

Lab Activities

Guidance (Lab Activity 1): prepare a Word file about the importance of email and precautions for email; make an email account; compose an email to send to your teacher about the assignment submission; attach the Word file with the email; send the email. The file content and the teacher's address are the student's own.
Guidance (Lab Activity 2, Project): create email accounts on two different websites, compare the differences and similarities, prepare a presentation on your work for different email accounts and present it in the class. These are hands-on tasks with no single written answer in the book. The unit's precautions for email use are: keep your password to yourself only and write it in some safe place; open email messages only from people you know; never

share your personal information with anyone on the internet; never make money payments without your parents' consent; never email your and your family pictures to strangers; avoid emails against religion and politics.

Unit 5: Algorithmic Thinking and Problem Solving

Exercise (pages 90-91): Tick the correct choice for the following statements/questions.

i. Which of the symbol is used for the flow of the program?

- input
- process
- decision
- ✓ arrows

ii. Which flow chart symbol is used for decision?

- ✓ diamond box
- rectangle
- arrows
- oval

iii. A set of instructions in simple language is called _____.

- flowchart
- ✓ algorithm
- compiler
- program

iv. Which of the shape of the flowchart symbols is like a diamond

- output
- input

- ✓ decision
- connector

v. Which statement is used to test a condition?

- print
- test
- loop
- ✓ if...then

vi. The condition is said to be _____ if the condition is met.

- False
- ✓ True
- equal
- not equal

vii. The first approach to solve a problem is _____.

- flowchart
- decision tree
- program
- ✓ algorithm

viii. Flowchart is a _____ of the solution of a problem.

- simple English
- ✓ pictorial
- technique
- first method

ix. In programming structure, we use _____ to repeat the

statements.

- if
- ✓ loop
- if...else
- if...then

x. _____ is a sequence of steps to solve a complex problem.

- ✓ Algorithm
- loop
- flowchart
- program

Exercise: Fill in the blanks.

i. A problem can be broken down into _____.

smaller problems

ii. _____ is breaking down a big problem into the smaller problems.

Decomposition

iii. A _____ is a programming structure that repeats a sequence of instructions until a specific condition is True.

loop

iv. Checking of condition is called _____ the condition.

testing

v. _____ is a graphical representation of the solution of

problem.

Flowchart

Exercise: Define the following terms.

i. Flowchart

Flowchart is a diagrammatic/graphical representation of a sequence of steps to solve a problem. A flowchart uses different standard symbols to design a solution to a problem; each symbol is connected with arrows to show the flow of the processing. The standard symbols are: Oval for Start and End; Parallelogram for Input and Output operations; Rectangle for processing, arithmetic operations and data-manipulations; Diamond for decision making, where there are two or three alternatives, true and false; Arrows (flow lines) to indicate the flow of logic by connecting symbols; Circle to replace long lines on a flowchart page; Off-page Connector when the target is on another page; Subroutine Symbol for a predefined process or function; Preparation for preprocessor or loops; and Comments for added explanation.

ii. Algorithm

An algorithm is a sequence of steps to solve a particular problem. The word 'algorithm' relates to the name of the Muslim mathematician Muhammad Ibne Musa Al-Khwarizmi, which means a procedure or a technique. Software Engineers commonly use an algorithm for planning and solving problems. An algorithm has the characteristics: Input (an algorithm may or may not require an input); Output (each algorithm is expected to produce at least one result); Definiteness (each instruction must be clear); Finiteness

(the algorithm should terminate after a finite number of steps).

iii. Selection

Selection is choosing between two actions when performing a task. In our daily life we break activities into steps or tasks, and in each task we need to make a selection. For example, while doing homework, after completing one task we check whether we have answered all the questions: if a question remains unanswered, we attempt it, and we check again; each time we check whether we have answered all the questions we are making a selection. We have to select from two options, for example: (1) Pack up our books. (2) Keep doing homework.

iv. If...Then

An If...Then statement consists of two parts: in the first part the condition is tested, and if the condition is true, then execute the actions in the 'then' part. The unit's example for completing homework is: (1) Check homework diary. (2) Start homework. (3) Answer the first quest. (4) IF all questions have been answered THEN start packing up. (5) If all the questions have not been answered, answer the next homework question.

v. Condition

A condition is a necessary requirement for action. We can write the condition in the form of a question, for example 'Have All the Questions been Answered?' The answer to this question may be Yes or No: if the answer is Yes, we pack up our books for the next day, else we continue with the homework. In programming terms, we

say that if the condition is true then 'do this' else 'do that'. Checking whether a condition is true or false is called testing the condition. The condition is said to be True if the condition is met.

Exercise: Write short answers to the following questions.

i. What is the IF condition?

An IF condition is the test part of an If...Then statement. A condition is a necessary requirement for action, written in the form of a question, for example 'Have All the Questions been Answered?' The answer may be Yes or No. If the condition is true, then the actions in the 'then' part are executed; checking whether a condition is true or false is called testing the condition.

ii. What is the difference between if...then and if...then...else statements?

An If...Then statement consists of two parts: in the first part the condition is tested, and if the condition is true, then execute the actions in the 'then' part. An If...Then...Else statement consists of three parts: first, in the If part it tells us what the condition is; second, the Then part tells us what to do if the condition is True; and third, the Else part tells us what to do if the condition is false. The unit's example is: (1) Have a look outside. (2) If there is still daylight, THEN ride a bicycle, ELSE clean your room. (3) Have a dinner.

iii. Write an algorithm using an IF...THEN statement for going to school in the morning by checking the weather outside.

Algorithm for going to school in the morning by checking the

weather outside, using an IF...THEN statement, in the unit's style: Step 1: Start. Step 2: Check the weather outside. Step 3: IF the weather is fine THEN get ready and go to school. Step 4: IF it is raining THEN take a raincoat or umbrella and go to school. Step 5: Stop. Flowchart: Oval (Start) joined by an arrow to a Parallelogram (Check the weather outside), then to a Diamond (Is the weather fine?), with the Yes branch going to a Parallelogram (Go to school) and the No branch going to a Parallelogram (Take a raincoat or umbrella and go to school), both joining to an Oval (Stop).

iv. Create an algorithm and flowchart for deciding whether to buy a laptop or a PC.

Algorithm for deciding whether to buy a laptop or a PC, using an IF...THEN...ELSE statement: Step 1: Start. Step 2: Decide whether you need to carry the computer to different places. Step 3: IF you need to carry the computer THEN buy a laptop ELSE buy a PC. Step 4: Stop. Flowchart: Oval (Start), arrow to Parallelogram (Decide: do you need to carry the computer?), arrow to Diamond (Need to carry the computer?), Yes branch to a Parallelogram (Buy a laptop), No branch to a Parallelogram (Buy a PC), both branches joining to an Oval (Stop). The choice follows the unit's rule that the Then part is what to do if the condition is True and the Else part is what to do if the condition is false.

v. Create an algorithm and flowchart for a table of 5.

Algorithm for a table of 5, using a loop: Step 1: Start. Step 2: Set COUNT = 1. Step 3: IF COUNT is greater than 10 THEN go to Step 7. Step 4: Print 5 x COUNT = 5 times COUNT. Step 5: Increase COUNT

by 1. Step 6: Go to Step 3. Step 7: Stop. This prints 5, 10, 15, 20, 25, 30, 35, 40, 45, 50. Flowchart: Oval (Start), arrow to Rectangle (COUNT = 1), arrow to Diamond (Is COUNT less than or equal to 10?), Yes branch to a Parallelogram (Print 5 x COUNT), then to a Rectangle (COUNT = COUNT + 1), with an arrow back to the Diamond; the No branch goes to an Oval (Stop). A loop is a programming structure that repeats a sequence of instructions until a specific condition is True, which is why the printing repeats until COUNT passes 10.

vi. Create an algorithm and flowchart to solve a quadratic equation. Algorithm to solve a quadratic equation, following the unit's own decomposition of the root formula x = (-b plus or minus square root of (b squared minus 4ac)) divided by 2a: Step 1: Start. Step 2: Input a, b and c. Step 3: Compute b squared. Step 4: Compute 4 times a times c. Step 5: Compute b squared minus 4 times a times c. Step 6: Compute the square root of (b squared minus 4 times a times c). Step 7: Compute -b plus or minus the square root, giving the two values. Step 8: Divide each value by 2a to get the two roots. Step 9: Print the two roots. Step 10: Stop. Flowchart: Oval (Start), arrow to Parallelogram (Input a, b, c), arrow to Rectangle (D = b squared minus 4ac), arrow to Rectangle (ROOT1 = (-b + square root of D) divided by 2a; ROOT2 = (-b - square root of D) divided by 2a), arrow to Parallelogram (Print ROOT1 and ROOT2), arrow to Oval (Stop). Note: the book's decomposition works out the square root step directly and does not discuss the case where b squared minus 4ac is negative, so this algorithm is stated as the book gives it, for the case where the square root exists.

Lab Activities

Guidance (Lab Activity 1): go to your lab and observe the steps to enter, work and come out from your lab, and write them as an algorithm. Guidance (Lab Activity 2): create an algorithm to open MS Word, create an application and save it on the computer. Guidance (Lab Activity 3): write an algorithm and flowchart to print an MS Word file. Guidance (Lab Activity 4): write an algorithm to take an admission in your school. These are hands-on and student-specific tasks; the book fixes no single written answer for them, so they are recorded as guidance only, in the unit's Start-to-Stop step style.

Unit 6: Binary Computing

Exercise Q1 (page 102): Tick the correct choice for the following statements/questions.

i. There are _____ numbers in binary system.

- ✓ 2
- 8
- 10
- 16

ii. One byte is a combination of _____ bits.

- 2
- ✓ 8
- 10
- 16

iii. Decimal system has _____ numbers.

- 2
- 8
- ✓ 10
- 16

iv. Number "17" is equal to _____ in binary system.

- 10000
- 10110

✓ 10001

10100

v. 1 Gigabyte is equal to _____.

✓ (1,024) cubed bytes

(1,024) to the power 4 bytes

(1,024) to the power 5 bytes

(1,024) to the power 6 bytes

vi. Hexadecimal system has total _____ numbers.

17

✓ 16

18

15

vii. Which number system consists of both digits and letters?

Binary

Octal

Decimal

✓ Hexa Decimal

viii. ASCII codes are of _____ bits.

4

✓ 7

8

16

ix. In binary addition 1 + 0 = _____.

0

✓ 1

10

11

x. Computer process data in _____ form.

✓ binary

decimal

octal

hexadecimal

Exercise Q2 (pages 102-103): Fill in the blanks.

i. Data in the form of digits is called _____ data.

digital

ii. In the binary system, 7 is equal to _____.

111

iii. Computer stores data in the _____ form.

binary

iv. The method to convert decimal number into binary is _____ method.

Division

v. BCD is _____ bit code.

4

Exercise Q3 (page 103): Answer the following questions.

i. Define Binary system.

The binary number system, also called the Base: 2 system, uses two digits (0, 1), in which each position represents a specific power of the base 2. It is a positional notation numbering system with base 2, with two values 0 and 1, and both digits' positions represent a power of two. Computers, calculators and all digital devices use the binary number system to store data and to solve problems. A computer is an electronic, digital device that stores, retrieves and processes data, usually in the binary number system; computers cannot understand words or numbers like humans, so everything is represented by a binary electrical signal of two states, ON (1) or OFF (0).

ii. Define Decimal System.

The decimal number system, also called the Base: 10 system, is the common number system that we use in our daily life routine, using digits from 0 to 9 with base 10. There are ten possible values in the base 10 system (0, 1, 2, 3, 4, 5, 6, 7, 8 and 9), each representing a specific power of the base 10. It is also known as the Hindu-Arabic or Arabic number system. In the decimal number system each digit is multiplied by an appropriate power of 10, depending on its position in the number; in a positional notation system the number base is called the radix, so the base ten system has a radix of 10.

iii. What is a bit?

A bit is the smallest unit of data in computing, which can be either 0 or 1. The term Bit is derived from Binary Digit, and the smallest

unit of memory is called a bit. Each On and Off value is called a bit: 0 means absence of electronic charge and 1 means presence of electric charge. A group of 8 bits is a Byte, a group of 4 bits (half of a byte) is a Nibble, and a group of bytes that a computer processes at one time is a Word.

iv. Define ASCII.

ASCII stands for American Standard Code for Information Interchange. It is the most widely used coding scheme. It is a 7-bit code, which means that only 128 characters, that is 2 to the power 7, can be represented. All characters on the keyboard are assigned a special code called ASCII. For example, if we want to store 'ALI', the computer needs 3 bytes, one character taking 1 byte in the memory: A's ASCII code is 65 and its binary code is 0100 0001; L's ASCII code is 76 and its binary code is 0100 1100; I's ASCII code is 73 and its binary code is 0100 1001.

v (a). Convert the decimal number 24 into its binary equivalent.

Divide the number by 2 and keep track of the remainders: 24 divided by 2 gives quotient 12, remainder 0.

Divide the quotient by 2: 12 divided by 2 gives quotient 6, remainder 0.

Divide the quotient by 2: 6 divided by 2 gives quotient 3, remainder 0.

Divide the quotient by 2: 3 divided by 2 gives quotient 1, remainder

1.

Divide the quotient by 2: 1 divided by 2 gives quotient 0, remainder 1.

Repeat the process with dividend equal to quotient till zero (0), as the unit teaches.

Read the remainders from bottom to top (MSB to LSB): 1, 1, 0, 0, 0.

(11000) base 2

v (b). Convert the decimal number 41 into its binary equivalent.
41 divided by 2 gives quotient 20, remainder 1.

20 divided by 2 gives quotient 10, remainder 0.

10 divided by 2 gives quotient 5, remainder 0.

5 divided by 2 gives quotient 2, remainder 1.

2 divided by 2 gives quotient 1, remainder 0.

1 divided by 2 gives quotient 0, remainder 1.

Read the remainders from bottom to top: 1, 0, 1, 0, 0, 1.

(101001) base 2

v (c). Convert the decimal number 101 into its binary equivalent.

101 divided by 2 gives quotient 50, remainder 1.

50 divided by 2 gives quotient 25, remainder 0.

25 divided by 2 gives quotient 12, remainder 1.

12 divided by 2 gives quotient 6, remainder 0.

6 divided by 2 gives quotient 3, remainder 0.

3 divided by 2 gives quotient 1, remainder 1.

1 divided by 2 gives quotient 0, remainder 1.

Read the remainders from bottom to top: 1, 1, 0, 0, 1, 0, 1.

(1100101) base 2

vi (a). Convert the binary number 110010 into its decimal equivalent.

Use the Multiplication method: each binary digit is multiplied by an appropriate power of 2, based on its position in the number.

From the binary point to the left, start with power of 2 as 0, 1, 2, 3 and so on.

110010 = 1 x 2 to the power 5 + 1 x 2 to the power 4 + 0 x 2 to the power 3 + 0 x 2 to the power 2 + 1 x 2 to the power 1 + 0 x 2 to the power 0.

= 1 x 32 + 1 x 16 + 0 x 8 + 0 x 4 + 1 x 2 + 0 x 1.

= 32 + 16 + 0 + 0 + 2 + 0.

= 50.

50

vi (b). Convert the binary number 10101 into its decimal equivalent.

10101 = 1 x 2 to the power 4 + 0 x 2 to the power 3 + 1 x 2 to the power 2 + 0 x 2 to the power 1 + 1 x 2 to the power 0.

= 1 x 16 + 0 x 8 + 1 x 4 + 0 x 2 + 1 x 1.

= 16 + 0 + 4 + 0 + 1.

= 21.

21

vi (c). Convert the binary number 1100100 into its decimal equivalent.

1100100 = 1 x 2 to the power 6 + 1 x 2 to the power 5 + 0 x 2 to

the power 4 + 0 x 2 to the power 3 + 1 x 2 to the power 2 + 0 x 2 to the power 1 + 0 x 2 to the power 0.

= 1 x 64 + 1 x 32 + 0 x 16 + 0 x 8 + 1 x 4 + 0 x 2 + 0 x 1.

= 64 + 32 + 0 + 0 + 4 + 0 + 0.

= 100.

100

vii (a). Perform the binary arithmetic operation: 1011 + 1110.

Write the numbers in columns, ones under ones: 1011 over 1110.

Ones column: 1 + 0 = 1. Write 1.

Twos column: 1 + 1 = 10 in binary. Write 0, carry 1.

Fours column: 0 + 1 + carried 1 = 10 in binary. Write 0, carry 1.

Eights column: 1 + 1 + carried 1 = 11 in binary. Write 1, carry 1.

Write the final carry 1 as a new leftmost bit.

Result: 11001. Check in decimal: 1011 is 11 and 1110 is 14; 11 + 14 = 25, and 11001 is 16 + 8 + 0 + 0 + 1 = 25.

11001

vii (b). Perform the binary arithmetic operation: 1110001 - 11111.

Write the numbers in columns with equal length: 1110001 over 0011111.

Ones column: 1 - 1 = 0.

Twos column: 0 - 1 cannot be done, so borrow 1 from the next column; 2 - 1 = 1. Borrow 1 is owed to the next column.

Fours column: 0 - 1 - borrowed 1 cannot be done, so borrow again; 2 - 1 - 1 = 0. Borrow 1 is owed.

Eights column: 0 - 1 - borrowed 1 cannot be done, so borrow again; 2 - 1 - 1 = 0. Borrow 1 is owed.

Sixteens column: 1 - 1 - borrowed 1 cannot be done, so borrow again; 2 - 1 - 1 = 1, because the borrowed 1 adds 2 to this column and the borrow is then repaid from the next column. Borrow 1 is owed.

Thirty-twos column: 1 - 0 - borrowed 1 = 0. No borrow remains.

Sixty-fours column: 1 - 0 = 1.

Result: 1010010. Check in decimal: 1110001 is 113 and 11111 is 31; 113 - 31 = 82, and 1010010 is 64 + 16 + 2 = 82.

1010010

vii (c). Perform the binary arithmetic operation: 1111 + 1000 + 1001.

Write the three numbers in columns: 1111 over 1000 over 1001.

Ones column: 1 + 0 + 1 = 10 in binary. Write 0, carry 1.

Twos column: 1 + 0 + 0 + carried 1 = 10 in binary. Write 0, carry 1.

Fours column: 1 + 0 + 0 + carried 1 = 10 in binary. Write 0, carry 1.

Eights column: 1 + 1 + 0 + carried 1 = 11 in binary. Write 1, carry 1.

Write the final carry 1 as a new leftmost bit.

Result: 100000. Check in decimal: 15 + 8 + 9 = 32, and 100000 is 32.

100000

viii. Write down the process of binary addition.

Binary addition is performed column by column from the rightmost (least significant) bit, exactly as the unit's worked examples add with carries. The rules are: 0 + 0 = 0; 0 + 1 = 1; 1 + 0 = 1; 1 + 1 = 10 in binary, so write 0 and carry 1 to the next column; 1 + 1 + carried 1 = 11 in binary, so write 1 and carry 1. Each carry is added

into the next column to the left, and a final carry becomes a new leftmost bit, as in part vii (a) above where 1011 + 1110 gives 11001.

ix. Describe the process of binary subtraction.

Binary subtraction is performed column by column from the rightmost bit, with borrowing, as in the unit's subtraction example. The rules are: 0 - 0 = 0; 1 - 0 = 1; 1 - 1 = 0; 0 - 1 cannot be done in that column, so borrow 1 from the next higher column, which adds 2 to the present column, giving 2 - 1 = 1, and the borrowed 1 is then subtracted from the next column. When the next column is also 0, the borrow is carried further left until a 1 is found, as in part vii (b) above where 1110001 - 11111 gives 1010010.

x. Convert the (ABC) base 16 into decimal.

In the hexadecimal system, digits used are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F, where A = 10, B = 11, C = 12, D = 13, E = 14 and F = 15.

Each hexadecimal digit is multiplied by an appropriate power of 16, based on its position in the number; from the hexadecimal point to the left, start with power of 16 as 0, 1, 2 and so on.

(ABC) base 16 = A x 16 to the power 2 + B x 16 to the power 1 + C x 16 to the power 0.

= 10 x 256 + 11 x 16 + 12 x 1.

= 2560 + 176 + 12.

= 2748.

2748

Lab Activities

Guidance (Lab Activity): make a chart for all 'A to Z' and 'a to z' with their ASCII values. The book gives the starting values: the ASCII value of A is 65 and of a is 97, so the chart runs A = 65, B = 66, C = 67, D = 68, E = 69, F = 70, G = 71, H = 72, I = 73, J = 74, K = 75, L = 76, M = 77, N = 78, O = 79, P = 80, Q = 81, R = 82, S = 83, T = 84, U = 85, V = 86, W = 87, X = 88, Y = 89, Z = 90; and a = 97, b = 98, c = 99, d = 100, e = 101, f = 102, g = 103, h = 104, i = 105, j = 106, k = 107, l = 108, m = 109, n = 110, o = 111, p = 112, q = 113, r = 114, s = 115, t = 116, u = 117, v = 118, w = 119, x = 120, y = 121, z = 122. Each letter takes the next value in order, which is how the unit's own examples run (A = 65, L = 76, I = 73).

Unit 7: Scratch Programming

Exercise Q1 (pages 114-115): Tick the correct choice for the following statements/questions.

i. _____ is a block to replay an action for a specific number of times?

- ✓ Repeat
- CAP
- Green Flag
- Hide

ii. _____ block is used to end the script.

- Repeat
- ✓ CAP
- Green Flag
- Hide

iii. _____ block is used to start the project.

- Repeat
- CAP
- ✓ Green Flag
- Hide

iv. _____ makes the sprite disappear from the stage.

- Repeat
- CAP

Green Flag

✓ Hide

v. _____ block controls the movements of the sprite.

✓ Motion

Arrow

Shrink

Stop

vi. Circular _____ enables full rotation of the sprite.

Motion

✓ Arrow

Shrink

Stop

vii. _____ button decreases the size of the sprite.

Motion

Arrow

✓ Shrink

Stop

viii. _____ button is used to stop the project at the time of execution.

Motion

Arrow

Shrink

✓ Stop

ix. _____ is the extension of the Scratch file.

✓ .sb2

.sc2

.scr

.sr2

x. The main working area in Scratch is called _____.

sprites

✓ stage

script

direction

Exercise Q2 (page 115): Fill in the blanks.

i. Scratch was created by the _____ group at MIT Media Lab.

Lifelong Kindergarten

ii. _____ deals with creating a script with the help of blocks.

Scratch Programming

iii. The _____ button allows you to decrease the size of the sprite.

Shrink

iv. The _____ tool is used to create a copy of an object.

Stamp

Note on fill iv: the unit gives no dedicated sentence naming the copy tool; the answer follows the unit's own Pen category, whose

stamp block stamps a copy of the sprite's image on the stage, and the editor's stamp cursor tool. This fill is therefore recorded as a best-reading, not as a sentence quoted from the unit.

v. A _____ is an object or character which you see on stage.

sprite

Exercise Q3 (page 115): Answer the following questions.

i. What is scratch?

Scratch was created by the Lifelong Kindergarten group at MIT Media Lab (Massachusetts Institute of Technology) for educational purposes. It is a visual programming language that provides an opportunity to students and even beginners to create their own interactive stories, games and animations. Scratch can be used by people of all ages, but it is primarily designed for ages 8 to 16 years students; they can design their own Scratch projects, and they learn to think creatively, reason systematically, and work collaboratively. Scratch can be freely downloaded at http://scratch.mit.edu, and once downloaded to a computer it does not need Internet access to create a project. The unit's summary also notes that Scratch was designed by Mitchel Resnick.

ii. What are blocks?

Blocks are the programming commands in the Blocks Area, the section that contains the various programming blocks or commands that can be used to build the program. The block categories shown in the unit are Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables and My Blocks. Blocks are dragged into the Scripts Area, the main coding area where the program logic is

developed. A script is the collection of stepwise instructions given to a sprite, built by snapping blocks together; a Hat Block is used to start a script, for example 'when green flag clicked'.

iii. On which tab do you Click to write a script?

Click on the Scripts tab to write a script. The unit's own project instructions say 'Go to Script tab', and its Figure 7.4 shows the Scripts tab above the Scripts Area. In the unit's later figures, which use the newer Scratch interface, the same tab is labelled Code, with Costumes and Sounds beside it.

iv. In which category is the block touching CAT present?

The touching block is in the Sensing category. The unit's block palette lists Sensing among the categories, and its sprite-move example uses a 'touching' test from the Sensing blocks, for example the script in Figure 7.1 which tests 'touching' and then points towards and says. Sensing blocks let a sprite detect what is around it, such as touching another sprite, the mouse-pointer or an edge.

v. Which sprite is by default on the scratch?

The default sprite in Scratch is the Scratch Cat. The unit's sprite-move program says: 'The default, Scratch Cat, will be created', and its Figure 7.4 shows the cat as Sprite1 on the stage.

Exercise Q4: Describe sprites. How do you add a new Sprite? Give Steps.

Q4. Describe sprites. How do you add a new Sprite? Give Steps.

In Scratch, the objects are known as Sprites. A sprite is an object or character which you see on the stage; the Stage is the area where the animated characters or objects are displayed and their behaviours are implemented, and the Sprite List is the list of animated characters or objects that are part of the project. Each sprite has its own scripting area. To add a new sprite, following the unit's own project steps: (1) In the new sprite panel, choose a sprite from the library, for example choose the bat sprite from the library and place it on the stage; or choose 'Upload sprite from file' and change its name to the name you want by clicking on the sprite. (2) Go to the Costumes tab and use the shrink option above the painting board to reduce its size if needed. (3) You can drag the sprite on the stage and place it where you want it, for example on the ground. (4) Click on the sprite in the Sprite List so that its own Scripts Area opens, and build its script there. The unit's two-sprite project adds a Cat and a Bat this way, each with its own script.

Exercise Q5: Write the steps to add sound in your script.

Q5. Write the steps to add sound in your script.

Steps to add sound in your script, using the unit's own interface: (1) Select the sprite you want to give a sound to, so its tabs open. (2) Click on the Sounds tab beside the Code and Costumes tabs. (3) Add a sound for the sprite from the sound library. (4) Go back to the Code (Scripts) tab, open the Sound category in the Blocks Area, and drag the 'play sound' block into the Scripts Area, placing it in your script at the point where the sound should play. The unit's own example scripts use the 'play sound' block, and its block palette lists Sound among the categories. Note: the unit shows the Sounds

tab and the Sound blocks but gives no separately numbered sound procedure, so these steps follow the unit's interface and its example scripts rather than a quoted procedure.

Exercise Q6: What is a variable? How do you create it? Give steps.

Q6. What is a variable? How do you create it? Give steps.

A variable is a named value that a program can store and change while it runs. The unit introduces variables through the countdown timer: 'In order to make a timer in Scratch, we'll need to use a variable.' The timer variable, named Time Remaining, keeps track of the amount of time you have left; it starts at 60 seconds and decreases by 1 every second until it reaches 0. The unit's learning outcomes likewise describe creating, setting and changing a variable under certain conditions, to either increase a value (for example, increase the score when the player touches a sprite) or decrease a value (for example, a countdown timer). To create a variable, following the unit's steps: (1) Navigate to the Variables section of the block menu. (2) Click the 'Make a Variable' button, which takes you to a new screen to create your new variable. (3) You will be prompted to enter a new variable name; enter a descriptive name, for example Time Remaining, and make sure that the 'For all sprites' option is selected, so that all sprites in your project can access the variable. (4) Click OK. (5) After creating the variable, check the box to the left of the variable in the block menu; this tells Scratch to display the variable on the stage, making it visible to the player. To use it as a timer, write on any sprite: when green flag clicked; set Time Remaining to 60; repeat 60 times: wait 1 second, change Time Remaining by -1; then say 'Game Over!' for

2 seconds and stop all.

Lab Activities

Guidance (Lab Activity, The Guessing Game): the game needs a game host sprite. The sprite will tell the user that it is thinking of a number between 1 and 100. The user is then given as many guesses as required to find the number the host is thinking about. To narrow the guesses, the host will tell the user 'too high' or 'too low' when the user guesses wrong. When the user guesses the right number, change the sprite's costume to reflect the user having won the game. Since the user must guess until they find the correct number, you will need to use a loop, and you will need two variables: one variable will hold the random number the host is thinking about (you might call it Host Number), and the other is for the user's guess, the answer variable supplied by Scratch; remember to hide the variable. Assemble the parts of the program in the host sprite. This is a hands-on project with no single fixed script printed in the book, so it is recorded as guidance following the unit's own hint.

Unit 8: Digital Citizenship

Exercise Q1 (pages 134-135): Tick the correct choice for the following statements/questions.

i. _____ is when someone copies your work.

- Copyright
- ✓ Plagiarism
- Phishing
- Piracy

ii. _____ is a cybercrime in which personal information is getting by email or phones.

- Copyright
- Plagiarism
- ✓ Phishing
- Internet

iii. _____ is the largest network in the word.

- LAN
- MAN
- Intranet
- ✓ Internet

iv. A _____ is media takes one side of the parties.

- digital citizenship
- cyberbullying

- ✓ bias
- defender

v. _____ must be installed in the computer to save from virus.

- ✓ antivirus
- program
- windows
- All of these

vi. It is necessary to have regularly change our _____.

- OS
- account
- ✓ password
- All of these

vii. ATM card number is an example of _____.

- ✓ confidentiality
- security
- Quality Assurance
- All of these

viii. Social hacking is also known as _____.

- social media
- social work
- ✓ social engineering
- business plan

ix. _____ is the main source of cyberbullying

confidentiality

security

✓ email

phone

x. You should not open emails and _____ from suspicious source.

files

✓ attachments

folders

applications

Note on MCQ ix: the unit does not name a single 'main source' of cyberbullying; it lists five channels where cyberbullying happens: Social Media first, then Messaging Apps, Online Games, Email, and Forums and Websites. Among the MCQ's options, email is the only channel the unit lists, so email is recorded as the best fit, not as a sentence quoted from the unit.

Exercise Q2 (page 135): Fill in the blanks.

i. _____ is known as information ethics.

Digital ethics

ii. _____ are the norms dedicated to ensuring the respect of others.

Digital ethics

Note on fills i and ii: the unit's own summary sentence covers both wordings: 'Digital ethics, also known as information ethics, are the

standards that ensure respect for users on the internet.' Both fills are therefore answered Digital ethics, following that sentence.

iii. _____ is a form of legal protection that gives ownership rights.

Copyright

iv. _____ means using someone else's words as yours.

Plagiarism

v. _____ is when someone replicates a copy of work of others.

Piracy

Exercise Q3 (page 135): Define the followings.

i. Copyright

Copyright is a legal protection that protects the rights of an author, for their original work. No one can copy your work, even published or unpublished. So, it protects the author of the original work from unauthorized duplication, stolen, distribution or plagiarized, and from other use. In other words, copyright is a right to copy, if the author of the work allows you to copy or reproduce the work. Copyright is a form of legal protection that gives you ownership rights to your original works. Copyright exclusive rights are: the right to control duplication of the work; the right to control the spreading and sale of the work; the right to control the display of the work or perform publicly; and it helps to generate derivative works based on the original work. The unit's ethics rule is: no one should use anyone's creative work, which can include a song, a video, a movie clip, a piece of visual art, a photograph and other

creative works, without authorization or compensation of the owner.

ii. Plagiarism

Plagiarism means using someone else's words or ideas without properly crediting the original author. For example, copying text and pasting it into a new document, and submitting a full document that is not your own. The unit's ethics rule is: no one should use someone else's work without giving proper credit to the original owner or creator.

iii. Piracy

Digital piracy refers to the illegal copying or distribution of copyrighted material via the Internet. It negatively affects the creative industries, including software, film, publishing, music and gaming. Piracy is when someone replicates a copy of a work and sells it without permission. The unit's ethics rule is: no one should replicate another's creative work, software, songs, videos or books.

iv. Cyberbullying

Cyberbullying is the use of digital platforms and technology to harass, intimidate, or harm others. It can occur through various online channels, such as social media, emails, text messages, or websites. The unit lists where cyberbullying happens: (1) Social Media, platforms Facebook, Instagram, Twitter, TikTok, for example posting hurtful comments, spreading rumors, or sharing embarrassing photos. (2) Messaging Apps, platforms WhatsApp, Snapchat, Messenger, Telegram, for example sending threatening

or abusive messages, creating group chats to exclude or target someone. (3) Online Games, multiplayer games like Fortnite, Minecraft, or Call of Duty, for example harassing other players, sending mean messages, or using voice chat to insult or threaten others. (4) Email, platforms Gmail, Yahoo Mail, Outlook, for example sending bullying emails, threatening messages, or bombarding someone with unwanted emails. (5) Forums and Websites, platforms Reddit, online discussion boards, or comment sections of blogs and news sites, for example posting negative comments, spreading false information, or attacking individuals in discussions. The unit's summary adds: cyberbullying includes sending, posting or sharing negative harmful or mean content about someone else.

v. Internet

The Internet is a network of networks. It is the largest network in the world. All the computers and devices are connected with each other. It allows digital information to travel around the world. The internet connection or access is established by ISP (Internet Service Providers) through different devices, in the form of a Wi-Fi signal, mobile phone and public internet, hotspots, broadband, cable and similar types.

vi. Digital citizenship

Digital citizenship is the responsible practice of the knowledge and technology skills by the technology consumers, and the responsible use of the internet, and digital devices to participate with the society. The unit also defines it as the ability of the responsible and proper use of the internet, computers and digital devices, and as

the norms of appropriate and responsible behavior regarding technology use. It helps the teachers, technology leaders, and parents, students, children, and technology users learn how to use technology properly.

vii. Defender

A Defender, in the cyberbullying context, is a person who stands up for and supports the victim of cyberbullying, rather than joining the bully or standing by silently. Note: this is flagged, because Unit 8 as printed gives no definition of Defender; the definition here is the standard cyberbullying-role meaning, given so the exercise term is not left blank, and it is not presented as a sentence quoted from the unit.

viii. Phishing

Phishing is the most popular digital social engineering attack, and a cybercrime. It is a trick used to steal money, identity, and expose personal information like credit card numbers, bank information, or passwords. The phishers are individuals or illegal groups who pretend themselves to be from reputable organizations, friends and family. They communicate through email, telephone or text message and use true appearance, with fake e-mails to collect information from the users, or connect them to a fake web site, which contains a link to a phishing website for the information. Phishing is a cybercrime in which personal information is obtained by email or phones.

ix. Social Hacking

Social Engineering and Social Hacking are closely related concepts that involve manipulating people to gain unauthorized access to information or systems. Social engineering is an approach of using, and cheating a victim or a target to gain the control of a computer system, to take someone's personal and financial information. It focuses on psychological tactics to exploit human behavior. Example: an attacker sends an email pretending to be from a company's IT department, asking the victim to reset their password using a fake link. Social hacking is a subset of social engineering that specifically refers to using social interactions to gain unauthorized access to systems or data. It often involves blending into a social environment to gather information or gain trust. Example: an attacker pretends to be a delivery person to gain access to a secure building, where they can then collect sensitive information. Social hacking is also known as social engineering in the exercise's wording, because the two are the unit's closely related pair.

x. Digital Ethics

Digital ethics, also known as information ethics, are the standards that ensure respect for users on the internet. Ethics is a set of moral principles that govern the behavior of an individual or group of people. ICT Ethics deals with the procedure, values, and practice that govern the process of using ICT and its related disciplines without damaging the moral values of an individual or organization. Examples include intellectual property rights, privacy policies, online etiquette, etc. The unit's ICT Ethics aims are: ensuring the access to right information; ensuring the information integrity and

security; enforcing online etiquette; enforcing the right to privacy; and enforcing the copyrights of information.

Exercise Q4: Write down the steps to secure information privacy and confidentiality.

Q4. Write down the steps to secure information privacy and confidentiality.

The unit's steps to secure information privacy and confidentiality are: (1) Never give out your name, address, phone number, e-mail, password or school name to un-related people or strangers. (2) Never share a picture of yourself with strangers. (3) Never open e-mails, files, or web pages from unknown sources or persons. (4) Always check with your teacher or parent before downloading any information or application. (5) Never respond to the strangers online. (6) Tell your teacher or parents if something makes you uncomfortable, scared or confused. (7) Tell your teacher or parents if anyone online asks you to meet in person. (8) Never share your movement plans or location on social network. (9) Never enter an online chat room without an adult's presence. (10) Do not share details about your life with others online. (11) Avoid using abusive language while chatting on social media. (12) Do not buy or order anything online without parents' supervision. (13) Don't save personal information on public devices. The unit adds for email care: do not create, forward or share spam and also use a spam filter; do not send or attach confidential documents and sensitive data in emails; do not open unexpected attachments and links in emails containing unwanted material; do not access, copy or transmit another's messages without permission; do not share and

send electronic communications that contain religious or political beliefs and abusive language. Confidentiality of information involves restricting data access strictly to authorized personnel, and users have a responsibility to ensure they maintain secure access control systems, for example passwords. Online privacy, also known as internet privacy, refers to how much of your personal, financial, and browsing data remains private when you are online.

Exercise Q5: Identify positive and negative impact of social media.

Q5. Identify positive and negative impact of social media.

Social media is a collection of technologies and applications that allow individuals to communicate, exchange information, and share photos and videos with one another. Social media impacts how people behave and perceive one another, and it can affect behavior, both online and offline. The Positive Impact of Social Media, as listed by the unit, is: it improves the connectivity and network to share feelings and opinions; it is used for positive communication and offers the opportunities for skilled networking; it offers an outlet for creativity in education and enhances social awareness; it shapes the new business opportunities and marketing channels. The Negative Impact of Social Media, as listed by the unit, is: there are fake relationships with fake profiles and that causes social isolation; people spread misinformation and rumors mostly on social media that compromise productivity; addiction to social media causes eye strain, eye fatigue, poor sleep, anxiety and depression; cyberbullying and harassment is common over social media; superficial connections promote cultural violence. The unit

also notes that social media can have a negative impact on students, both physically and mentally.

Exercise Q6: What are the threats on the internet to your devices or data?

Q6. What are the threats on the internet to your devices or data?

The unit's main threats and how to secure ourselves from them are: (i) Deletion of Data Accidentally: deletion of data from the computer is very common; sometimes a good user can delete his own data accidentally. The two solutions are: proper rights must be given to the user by the administrator, and periodic backups of data should be taken to recover deleted files. (ii) Firewalls: the biggest problem with computers is that they are not always secure; to protect data from unauthorized access, we must use firewalls. A firewall is a security system designed to monitor and control incoming and outgoing network traffic based on predetermined security rules; it acts as a barrier between a trusted internal network and untrusted external networks, such as the internet. (iii) Virus: a virus is a type of malicious software designed to infect and damage computer systems by spreading from one file to another, often without the user's knowledge. It can corrupt data, steal information, or render a system inoperable. To protect against viruses, it is essential to install and regularly update antivirus software, avoid downloading files or clicking links from unknown sources, and keep your operating system and applications updated with the latest security patches. Additionally, backing up important data can help mitigate the impact of a virus infection. A computer virus is a kind of malicious computer program, which when executed, replicates

itself and inserts its own code; when the replication is done, this code infects the other files and programs present on your system. (iv) Natural Hazards: hardware (data storage) can be damaged by fire, floods, earthquake or falling of buildings, so it is important to save backups away from the computer room. (v) Hackers: hacking is the unauthorized access to or manipulation of computer systems, networks, or data. Hackers often exploit vulnerabilities in software or use techniques like phishing to steal information, disrupt services, or gain control of systems. To protect yourself from hacking: use strong and complex passwords and change and update them regularly; use Two-Factor Authentication (2FA) to add an additional layer of security to your accounts; regularly update your operating system, applications, and security software to patch vulnerabilities; avoid clicking on suspicious links or downloading attachments from unknown sources; avoid using public Wi-Fi for sensitive transactions, and use a VPN (Virtual Private Network) if necessary. Cyber Threats and Attacks in general include various malicious activities and tactics used to compromise digital security, including social engineering, phishing and cyberbullying as defined above. The unit also advises: keep your antivirus/antimalware software updated, make sure automatic updates are working or make it a habit to download the latest antivirus programs; do not open emails and attachments from suspicious sources; use multifactor authentication; and be wary of tempting offers, since if an offer sounds too enticing, think twice before accepting it as a fact.

Lab Activities

Guidance (Lab Activity 1): begin with a brief discussion or presentation on what Digital Citizenship is and why it is important, highlighting key aspects such as online safety, digital etiquette, protecting personal information, and the impact of social media. Guidance (Lab Activity 2): ask students to create a 'Digital Citizenship Guide' using word processing software; this guide should include tips and best practices for responsible online behavior, protecting personal information, and respecting others in digital spaces, and creativity may be encouraged by allowing images, bullet points, and headings. Guidance (Lab Activity 3): start an activity with a class discussion where students can share what they learned, any challenges they faced, and how they plan to apply these principles in their everyday online interactions. These are classroom activities with no single written answer in the book.

Unit 9: Entrepreneurship in Digital Age

Exercise Q1 (pages 146-147): Tick the correct choice for the following statements/questions.

i. _____ is the smart application of the power of digital technologies to business.

- ✓ Digital entrepreneurship
- vision
- technology
- strategy

ii. Which of the following is not component on digital entrepreneurship?

- ✓ Digital entrepreneurship
- vision
- technology
- strategy

iii. _____ is the creator of Facebook.

- ✓ Mark Zuckerberg
- Bill Gates
- Steve Jobs
- Travis Kalanick

iv. _____ introduced Uber.

- Mark Zuckerberg

Bill Gates

Steve Jobs

✓ Travis Kalanick

v. _____ uses social media, phones, and other electronic devices to advertise a product or service.

✓ Digital Promotion

proposition

Quality Assurance

All of these

vi. A _____ is a statement that clearly identifies the benefits of company to the customers.

Digital Promotion

✓ proposition

Quality Assurance

All of these

vii. _____ is the process of verifying whether the product or services meet the customers' expectations.

Digital Promotion

proposition

✓ Quality Assurance

All of these

viii. _____ is a simple sample of your future design, or a test case that demonstrates the finished product.

ideation

- ✓ prototype
- implementation
- test

ix. Bill gates and Paul Allen Co-founded _____, and it is now the world's prime market capitalization company.

- Amazon
- Uber
- Facebook
- ✓ Microsoft

x. _____ create the space for the end-users and identify the needs and demands of the target audience.

- entrepreneurship
- digital entrepreneurship
- ✓ design thinking
- digital technologies

Note on MCQ ii: the unit's principal components of Digital Entrepreneurship are the vision, the technology and the strategy; Digital entrepreneurship itself is the whole, not one of its components, so it is the option that is not a component.

Exercise Q2 (pages 147-148): Fill in the blanks.

i. _____ is the smart application of digital technologies.

Digital entrepreneurship

ii. Digital entrepreneurship is the business using _____.

digital technologies

iii. Netflix is an example of design _____.

thinking

iv. _____ uses social media, phones to advertise products.

Digital promotion

v. _____ is a human-centered approach.

Design Thinking

Exercise Q3 (page 148): Answer the following questions.

i. Define digital entrepreneurship.

Digital entrepreneurship is the smart application of the power of digital technologies to business. The unit defines it as the process of using digital technologies to start, manage and run the businesses. It uses the internet, mobile devices, social media and other digital tools to create, deliver, and scale products or services. Digital entrepreneurship includes: identifying opportunities to reach and find customers; designing and presenting new products and services; generating income, and reducing the cost of products and services; and cooperating with partners to manage risk and competitive advantage. The principal components in Digital Entrepreneurship are the vision, the technology and the strategy. It is not just a computer business: digital entrepreneurship can apply to different sectors of the economy, for example Agriculture, trading, manufacturing, education, real estate, etc., and it can apply both to an existing business and a new business.

ii. Define digital promotion.

Digital promotion uses social media, phones, and other electronic devices to advertise a product or service. This includes those fancy electronic billboards that are alongside highways. Digital promotion is the planning or overall scope of your plan to attract and retain customers. Digital advertising is any visual ad, written or video, that you see on the internet.

iii. Define value proposition.

A value proposition is a statement that clearly identifies the benefits a company's products and services will deliver to its customers. A company should view its value proposition statement as an opportunity to tell customers why the company, its product or its service is the best one to buy. It summarizes why the company can best meet the customer's needs or best solve the customer's problems.

iv. Define design thinking.

Design thinking is a non-linear, iterative process that teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test. Design thinking is involving five phases: Empathize, Define, Ideate, Prototype and Test. It is a human-centered approach: it creates the space for the end-users and identifies the needs and demands of the target audience, and it encourages a mind to move from just problem solving to identifying the opportunities for significant impact.

v. Define quality assurance.

Quality assurance is the process of verifying whether the product or services meet the customers' expectations. For example, Quality Assurance in Software Testing means a procedure to ensure the quality of software products or services provided to the customers by an organization. Quality assurance focuses on improving the software development process and making it efficient and effective as per the quality standards defined for software products.

Exercise Q. No 4: Describe digital entrepreneurship.

Q. No 4. Describe digital entrepreneurship.

Digital entrepreneurship is the smart application of the power of digital technologies to business. It is the process of using digital technologies to start, manage and run businesses, using the internet, mobile devices, social media and other digital tools to create, deliver, and scale products or services. It includes identifying opportunities to reach and find customers, designing and presenting new products and services, generating income and reducing the cost of products and services, and cooperating with partners to manage risk and competitive advantage. Its three principal components are the vision, the technology and the strategy. It is not just a computer business: it can apply to different sectors of the economy, for example Agriculture, trading, manufacturing, education, real estate, and to both an existing business and a new business. The unit's examples of digital entrepreneurs are: Bill Gates, who with Paul Allen co-founded Microsoft in 1975, now the world's prime market capitalization company; Jeff Bezos, who in 1994 founded Amazon.com with the idea of creating an online bookstore, which expanded to music,

DVDs, electronics, and more; Larry Page and Sergey Brin, who in 1996 created Google, which includes YouTube, Google Maps, and Gmail; Mark Zuckerberg, who in 2004 created Facebook when he was a 20-year-old student at Harvard University; Travis Kalanick and Garrett Camp, who in 2009 developed the world's famous online taxi business Uber; Tim Hwang, in 2013 founder and CEO of FiscalNote; Colin Walsh, in 2015 founder and CEO of Varo Bank; and Elon Musk, who cofounded companies including SpaceX created in 2002, X, previously known as Twitter, in 2022 and rebranded in July 2023, xAI in 2023, The Boring Company in 2016, Neuralink Corporation in 2024, OpenAI in 2015, Tesla in 2003, SolarCity in 2004 and PayPal in 1998. Entrepreneurship itself is an ability to establish and run a business enterprise; it is a fresh starting of a business with some doubts about profit, but the aim is making profit, finding opportunities, and taking risks to establish and grow a new business. An entrepreneur is an individual who creates a new business, bearing most of the risks and enjoying most of the rewards, commonly seen as an innovator, a source of new ideas, goods, services, and business or procedures. The characteristics of entrepreneurship are: Identifying Opportunities, identifying needs or problems that people have, which can be solved with new products or services; Innovative Solutions, creating new and unique ideas, products, or services that stand out and offer something different from what is currently available; Risk-Taking, investing time, money, and resources into a venture with the potential for both high rewards and high risks; Resource Management, efficiently managing resources, including finances, personnel, and technology, to achieve business goals; and Business Growth, scaling the

business to increase market reach, revenue, and impact.

Exercise Q. No 5: Describe the design thinking and explain its steps.

Q. No 5. Describe the design thinking and explain its steps.

Design thinking is a non-linear, iterative process that teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test. It is a human-centered approach, and it is more than just a process: it opens up an entirely new way to think, and it offers a collection of hands-on methods to help you apply this new mindset. Traditional problem solving involves coming up with one solution, but design thinking first makes us diverge, by trying to generate a diverse assortment of possible alternative solutions to a problem. Design thinking involves five phases, and understanding the five steps will give you confidence and enable you to use the technique in your study and job, and to address difficult issues that arise in organizations, nations, and all over the world. The five steps are:

(1) Empathize: research your users' needs. This is the first phase of design thinking, where you gain real insight into users and their needs. It focuses on user-centric research. It is important to develop an empathic understanding of the issue you are attempting to resolve. Consult specialists to learn more about the problem, and make observations to connect with and understand your users. You may also want to immerse yourself in your users' physical environment to gain a deeper, personal understanding of the issues involved, as well as their experiences and motivations. Empathy is essential to problem solving and a human-centered design process, as it allows design thinkers to set aside their own

assumptions about the world and gain real insight into users and their needs. (2) Define: state your users' needs and problems. This is the second phase, where you define the problem statement in a human-centered manner. In the Define stage, you will organize the information you have gathered during the Empathize stage, analyze your observations to define the core problems you and your team have identified up to this point, and define the problem and problem statement in a human-centered manner. (3) Ideate: challenge assumptions and create ideas. This is the third phase, where you identify innovative solutions to the problem statement you have created. During this stage, designers are ready to generate ideas: you have grown to understand your users and their needs in the Empathize stage, and you have analyzed your observations in the Define stage to create a user-centric problem statement, and with this solid background, you and your team members can start to look at the problem from different perspectives and ideate innovative solutions to your problem statement. (4) Prototype: start to create solutions. This is the fourth phase, where you identify the best possible solution. The design team will now produce a number of inexpensive, scaled-down versions of the product, or specific features found within the product, to investigate the key solutions generated in the ideation phase. These prototypes can be shared and tested within the team itself, in other departments or on a small group of people outside the design team. A prototype is a simple sample of your future design, or a test case that demonstrates the finished product. (5) Test: try your solutions out. This is the fifth and final stage of design thinking, which involves testing ideas to gain a thorough grasp of

the final product and its intended audience. Using the best options found in the Prototype stage, designers or evaluators extensively test the finished solution. This is the final stage of the five-stage model; however, in an iterative process such as design thinking, the results generated are often used to redefine one or more further problems. This deeper level of comprehension may help you to look into the situations of use and how users interact with the product, and it might even push you to go back to an earlier step in the design thinking process. You can then proceed with further iterations and make alterations and refinements to rule out alternative solutions. The ultimate goal is to get as deep an understanding of the product and its users as possible. The benefits of design thinking for entrepreneurs are: Customer-centric Innovation, a customer-centric approach confirms that the wants and likings of the customer are at the basic priority of the creative process; Repetitively Problem-solving, problem-solving allows entrepreneurs to test the ideas, gather responses, and solve the problem; Creativity and Innovation, entrepreneurs can explore new opportunities for innovation and differentiation in saturated markets; Collaboration and Authorization, it is a team work, bringing together the cross-functional teams with different viewpoints and skills to produce a strong solution; and Adaptability and Flexibility, entrepreneurs must have the quality to adapt and be flexible to look for the navigated doubts, and turn them when necessary.

Exercise Q. No 6: What is the example of design thinking, Explain in detail?

Q. No 6. What is the example of design thinking, Explain in detail?

The unit gives five real-world examples of design thinking in entrepreneurship. Airbnb: 'Airbnb' is an abbreviation of 'Air bed and Breakfast'. Airbnb developed the hospitality production by applying design thinking principles. They identified the pain points of travelers, tourists and explorers, and provided them unique, affordable accommodations. Airbnb provides the trust environment between hosts and guests, and focused on customer experience and reviews. In this type of service, the property owners provide their spaces to travelers on rent. Dropbox: Dropbox is a Service Software; it is cloud-based file storage to store, access, synchronize the content and share files and documents from anywhere safely. Dropbox recognized the frustration users were facing with traditional file storage and sharing ways, so they applied design thinking to develop a simple and natural solution. It is popular within individuals and businesses too. Tesla: Tesla is also an example of design thinking principles; it is an automotive industry that addresses the rising demand for sustainable transportation. By environmentally aware consumers and primary adopters of electric vehicles, Tesla knew the need for high-performance electric vehicles, like the Model S, Model 3, Model X, and Model Y, with inspiring range and advanced features. Netflix, explained in detail: many companies are successfully using design thinking, but Netflix proved to be the giant of the design thinking business. In 2011, Netflix took its design thinking ahead and responded to consumers' need for original, provoking content that was not airing on old networks. Later, in 2016, they added short trailers to the interface. Netflix is a good example of thinking process, because it began

from the users' need, empathized with viewers who wanted original content beyond old networks, defined that unmet need, ideated a new kind of service, prototyped it as a streaming interface, and tested and refined it, for example by adding short trailers to the interface in 2016. Nike: Nike's design ideology is moving forward; it is a favorite brand among athletes for nearly five decades. Their focus is creating new and groundbreaking productions for their consumers.

Exercise Q. No 7: What is meant by sustainable development? Give some examples.

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Sustainable development is development that meets the needs of the present, without compromising the ability of future generations to meet their own needs. The concept of sustainable development can be interpreted in many different ways, but at its core is an approach to development that looks to balance different, and often competing, needs against an awareness of the environmental, social and economic limitations we face as a society. The unit's diagram names five pillars: People, end poverty and hunger in all forms and ensure dignity and equality; Planet, protect our planet's natural resources and climate for future generations; Prosperity, ensure prosperous and fulfilling lives in harmony with nature; Partnership, implement the agenda through a solid global partnership; and Peace, foster peaceful, just and inclusive societies. The three types of sustainable development are: Economic viability, Environmental protection, and Social equity. The Sustainable

Development Goals are: to promote the kind of development that minimizes environmental problems, for example quality education, zero hunger, etc.; and to meet the needs of the existing generation without compromising the quality of the environment for future generations. The unit's examples of sustainable development are: wind energy, solar energy, green environment, sustainable construction, and sustainable forestry.

Exercise Q. No 8: How we can achieve the sustainable development goals?

Q. No 8. How we can achieve the sustainable development goals?

Sustainable development can be achieved if we follow the following points: it can be achieved by limiting human activities; technological development should be input effective and not input utilizing; the rate of consumption should not surpass the rate of recovery; all types of pollution should be minimized; and it can be achieved by sensible use of natural resources. In the unit's terms, achieving the goals means promoting the kind of development that minimizes environmental problems, for example quality education and zero hunger, and meeting the needs of the existing generation without compromising the quality of the environment for future generations.

Lab Activities

Guidance (Lab Activity 1): prepare a group presentation to present the topic 'Entrepreneurship in Pakistan' and present it in front of the class; teacher or parents may help the students to prepare the presentation. Guidance (Lab Activity 2): think about the 'Design

Thinking Process' and prove that it is extremely useful when you tackle problems that are unknown. Guidance (Lab Activity 3): prepare a chart related to 'Sustainable Development Goals'. These are classroom activities with no single written answer in the book.